

# SOC INVESTIGATION REPORT

Wazuh SOC Home Lab — Investigation Report

Prepared by Saqlain Abbas

Personal SOC Home Lab • Detection & Investigation

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|---------------|--------------------------------------------------------------|
| Analyst       | Saqlain Abbas                                                |
| Project       | Wazuh SOC Home Lab                                           |
| Environment   | Personal SOC Home Lab (VirtualBox)                           |
| Endpoints     | MYDFIR-Saqo (Windows agent), saqo-linux (Ubuntu/Linux agent) |
| Attacker host | Kali Linux (attacker VM)                                     |
| Report date   | September 12, 2026                                           |

Analyst note: This report is based on telemetry generated and investigated in my own Wazuh home lab. I distinguish confirmed evidence (from Wazuh Discover/Dashboard) from interpretation, and document limitations where the available data does not fully confirm intent.

# 1. Executive Summary

During this investigation I reviewed telemetry collected by my Wazuh manager from two active agents (MYDFIR-Saqo, a Windows endpoint, and saqo-linux, an Ubuntu endpoint), with a Kali Linux VM used as the attacker machine. The investigation identified Windows account-management activity on MYDFIR-Saqo, including the Guest account being enabled and a cluster of Windows account-change events (event IDs 4722, 4724, 4725, 4732, 4733, 4738), a set of scripting files created under the Windows Temp/User folder, and 12 failed Windows logon attempts recorded on the SOC activity dashboard. Separately, saqo-linux recorded repeated failed SSH authentication attempts consistent with a brute-force attempt. A custom Wazuh detection rule identified the repeated SSH failures, and Wazuh Active Response automatically blocked the source host using the firewall-drop script. The activity was intentionally generated as part of my own SOC home lab practice; the same sequence in a production environment would warrant investigation for unauthorized access and brute-force activity.

# 2. Investigation Scope & Methodology

## Scope

I reviewed Windows account-management events, Windows failed logon activity, scripting/file creation events, Linux SSH authentication logs, the custom Wazuh detection rule, and the Active Response event generated in my Wazuh lab, using the Discover, Endpoints, and Dashboard modules of the Wazuh interface.

## Method

I reviewed the events chronologically in Wazuh Discover and correlated them using timestamps, hostnames, event descriptions and rule descriptions. I used the Endpoints view to confirm which agents were active during the investigation window, and the SOC Activity Overview dashboard to quantify failed logons and account changes over time.

# 3. Findings

Each finding follows the structure: FACT → EVIDENCE → SECURITY SIGNIFICANCE → SCREENSHOT.

## 3.1 — Guest Account Enabled

**FACT:** The local Guest account on MYDFIR-Saqo was enabled. Wazuh recorded this as a Windows account-changed event on Sep 1, 2026 at 21:23:00 (local Wazuh time).

|           |                                              |
|-----------|----------------------------------------------|
| Host      | MYDFIR-Saqo                                  |
| Event     | Guest account enabled / User account changed |
| Timestamp | Sep 1, 2026, 21:23:00.634                    |
| Account   | Guest                                        |

**Security significance:** The Guest account is normally disabled in a secure Windows configuration. Unexpected activation increases the available attack surface and should be investigated.

**Screenshot:** Discover — wazuh-alerts-\* (guest account enabled).

## 3.2 — Windows Account Change Cluster

**FACT:** The SOC Activity Overview dashboard shows a cluster of Windows account-management events (event IDs 4722, 4724, 4725, 4732, 4733, 4738) occurring close together in time on MYDFIR-Saqo.

|        |                                                            |
|--------|------------------------------------------------------------|
| Host   | MYDFIR-Saqo                                                |
| Events | 4722, 4724, 4725, 4732, 4733, 4738                         |
| Window | Late Aug &ndash; early Sep 2026 (per dashboard time range) |

**Security significance:** This combination of event IDs typically spans account enablement, password/attribute changes, and group-membership changes. A tight cluster of these events on one host is a pattern worth reviewing for unauthorized account tampering or privilege escalation.

**Screenshot:** Dashboard — "Windows Account Changes Over Time" panel.

### 3.3 — Scripting Files Created Under Windows Temp/User Folder

**FACT:** Wazuh recorded four consecutive "Scripting file created under Windows Temp or User folder" alerts on MYDFIR-Saqo within a 30-millisecond-wide window.

|            |                                                          |
|------------|----------------------------------------------------------|
| Host       | MYDFIR-Saqo                                              |
| Rule       | Scripting file created under Windows Temp or User folder |
| Timestamps | Sep 1, 2026, 21:22:45.685 &ndash; 21:22:45.761           |

**Security significance:** Script files dropped into Temp or user-writable folders are a common technique for staging or executing payloads, and are worth correlating with the account-change activity that followed moments later.

**Screenshot:** Discover — wazuh-alerts-\* (scripting file creation events).

### 3.4 — Failed Windows Logon Activity

**FACT:** The SOC Activity Overview dashboard recorded 12 failed Windows logon events on MYDFIR-Saqo during the reporting window.

|        |                                        |
|--------|----------------------------------------|
| Host   | MYDFIR-Saqo                            |
| Metric | Failed Windows logon &ndash; Count: 12 |

**Security significance:** A double-digit count of failed logons on an endpoint is consistent with repeated incorrect credential attempts and should be reviewed alongside the account-change cluster above.

**Screenshot:** Dashboard — "Failed Windows logon" panel.

### 3.5 — SSH Brute-Force Activity Against saqo-linux

**FACT:** Wazuh recorded repeated failed SSH authentication attempts against saqo-linux, including sshd authentication failures, unix\_chkpwd password check failures, a syslog "missed the password more than one time" event, and an sshd connection reset, all within a five-second window.

|            |                                                 |
|------------|-------------------------------------------------|
| Host       | saqo-linux                                      |
| Source     | Kali Linux attacker VM                          |
| Timestamps | Sep 11, 2026, 15:32:24.640 &ndash; 15:32:30.701 |

**Security significance:** Repeated authentication failures from the same source in a short interval are consistent with automated password-guessing / brute-force activity.

**Screenshot:** Discover — wazuh-alerts-\* (SSH failure sequence).

### 3.6 — Custom Wazuh Detection

**FACT:** A custom Wazuh rule, described as "Multiple SSH login failures observed from the same source IP", fired at 15:32:30.665 after the repeated SSH failures against saqo-linux.

|                  |                                                              |
|------------------|--------------------------------------------------------------|
| Host             | saqo-linux                                                   |
| Rule description | Multiple SSH login failures observed from the same source IP |
| Timestamp        | Sep 11, 2026, 15:32:30.665                                   |

**Security significance:** This demonstrates that the custom detection successfully converted several low-level authentication failures into a single higher-confidence alert, reducing manual review effort.

**Screenshot:** Discover — wazuh-alerts-\* (custom rule triggered).

### 3.7 — Automated Response

**FACT:** Immediately after the custom detection fired, Wazuh Active Response executed a firewall-drop action, recorded as "Host Blocked by firewall-drop Active Response" at 15:32:30.944.

|               |                               |
|---------------|-------------------------------|
| Host          | saqo-linux                    |
| Rule / Action | firewall-drop Active Response |
| Timestamp     | Sep 11, 2026, 15:32:30.944    |

**Security significance:** The automated response reduced immediate risk by blocking further connection attempts from the attacking host within roughly 300 milliseconds of detection.

**Screenshot:** Discover — wazuh-alerts-\* (firewall-drop block event).

## 4. Endpoint & Agent Overview

At the time of this investigation, the Wazuh Endpoints view confirmed the lab's agent inventory:

| Metric           | Value                                                      |
|------------------|------------------------------------------------------------|
| Agents by status | Active: 2, Disconnected: 0, Pending: 0, Never connected: 0 |
| Top OS           | Kali: 1, Windows: 1                                        |
| Top groups       | default: 2                                                 |

Both monitored endpoints (Windows and Linux) were active and reporting throughout the investigation window, confirming that the telemetry gaps, if any, are not due to disconnected agents.

## 5. Investigation Summary — What Happened?

Correlating the Windows and Linux telemetry, the activity on MYDFIR-Saqo began with scripting files being created under the Temp/User folder, closely followed by the Guest account being enabled and a cluster of account-management events (4722/4724/4725/4732/4733/4738), alongside 12 recorded failed Windows logons. Separately, on Sep 11, 2026, saqo-linux recorded a rapid sequence of failed SSH authentication attempts originating from the Kali attacker VM. My custom Wazuh rule detected the repeated SSH failures within milliseconds, and the firewall-drop Active Response blocked the source host immediately afterward.

## 6. Incident Timeline

| # | Date         | Time         | Host        | Event                                                  |
|---|--------------|--------------|-------------|--------------------------------------------------------|
| 1 | Sep 1, 2026  | 21:22:45     | MYDFIR-Saqo | Scripting files created under Temp/User folder         |
| 2 | Sep 1, 2026  | 21:23:00     | MYDFIR-Saqo | Guest account enabled / account changed                |
| 3 | &ndash;      | &ndash;      | MYDFIR-Saqo | Account-change cluster (4722/4724/4725/4732/4733/4738) |
| 4 | &ndash;      | &ndash;      | MYDFIR-Saqo | 12 failed Windows logons recorded                      |
| 5 | Sep 11, 2026 | 15:32:24     | saqo-linux  | SSH authentication failures begin                      |
| 6 | Sep 11, 2026 | 15:32:30.665 | saqo-linux  | Custom rule: multiple SSH failures, same source        |
| 7 | Sep 11, 2026 | 15:32:30.944 | saqo-linux  | firewall-drop Active Response &mdash; host blocked     |

Times are as recorded by Wazuh Discover for each event group.

## 7. Who, What, When, Where, Why, How

**Who:** The account-management activity was associated with the local Guest and other Windows accounts on MYDFIR-Saqo. The SSH activity originated from the Kali Linux attacker VM and targeted saqo-linux.

**What:** The Guest account was enabled, a cluster of Windows account-change events occurred, scripting files were created under Temp/User folders, 12 failed Windows logons were recorded, and repeated SSH authentication failures were observed and blocked.

**When:** The Windows-side activity was recorded on Sep 1, 2026 around 21:22–21:23. The Linux-side brute-force and response occurred on Sep 11, 2026 between 15:32:24 and 15:32:30.

**Where:** MYDFIR-Saqo (Windows) was the source of the account and scripting activity. saqo-linux (Ubuntu) was the SSH target. The Kali VM acted as the attacker. Wazuh collected and correlated telemetry from both agents.

**Why:** The activity was intentionally generated by me as part of my own SOC home lab practice. In a production environment, this combination of account tampering and SSH brute-forcing would be treated as suspicious and investigated for unauthorized access.

**How:** The Windows-side changes were generated through local account actions on MYDFIR-Saqo. SSH brute-force attempts were launched from the Kali VM against saqo-linux. Wazuh collected the telemetry, my custom rule detected the repeated SSH failures, and firewall-drop automatically blocked the attacking host.

## 8. Detection & Response Assessment

### Detection

My custom Wazuh rule detected repeated SSH authentication failures from the same source within a short window, converting several low-level authentication events into one higher-confidence detection that was easier to investigate.

### Response

Following detection, Wazuh Active Response executed the firewall-drop script against the attacking host, successfully blocking further connection attempts, confirmed by the "Host Blocked by firewall-drop Active Response" event recorded roughly 300 milliseconds after the detection fired.

### Limitations

The telemetry confirms the observed actions but does not, by itself, prove malicious intent, since the activity was intentionally generated in a personal lab for practice. The Windows account-change cluster and the Linux SSH brute-force were not observed to be part of a single coordinated attack chain, and are reported as separate findings.

## 9. Recommendations

- **Guest account** — Keep the Guest account disabled by default and alert on unexpected enablement events.
- **Privileged account changes** — Monitor event IDs 4722/4724/4725/4732/4733/4738 as a group and investigate tight clusters of these events on a single host.
- **Scripting/temp folder monitoring** — Continue alerting on script files created under Temp or user-writable folders and correlate with subsequent account activity.

- **Failed logon monitoring** — Set a threshold-based alert for repeated failed Windows logons per host.
- **SSH hardening** — Prefer SSH key-based authentication, restrict SSH to trusted networks, and consider disabling direct root login.
- **Brute-force protection** — Keep Wazuh Active Response enabled and consider a layered OS-level control such as fail2ban.
- **Detection engineering** — Keep custom rules documented and version-controlled; re-test them after any Wazuh upgrade.
- **Active Response validation** — Periodically re-test firewall-drop to confirm the block is still effective and logged correctly.

## 10. Conclusion

This investigation reconstructed two related but separately-timed sets of activity in my Wazuh SOC home lab: an account-management and scripting-activity cluster on the Windows endpoint MYDFIR-Saqo, and an SSH brute-force attempt from a Kali attacker VM against the Linux endpoint saqo-linux. The SSH activity was detected by a custom Wazuh rule and automatically mitigated through firewall-drop Active Response within roughly 300 milliseconds. Although I intentionally generated this activity for practice, the same behavioral patterns in a production environment would warrant investigation for unauthorized access and account tampering.

## 11. Evidence & Screenshot Register

| # | Evidence                                            | Used in section |
|---|-----------------------------------------------------|-----------------|
| 1 | Discover &mdash; account changed / guest enabled    | Finding 3.1     |
| 2 | Dashboard &mdash; Windows Account Changes Over Time | Finding 3.2     |
| 3 | Discover &mdash; scripting file creation events     | Finding 3.3     |
| 4 | Dashboard &mdash; Failed Windows logon panel        | Finding 3.4     |
| 5 | Discover &mdash; SSH failure sequence               | Finding 3.5     |
| 6 | Discover &mdash; custom rule triggered              | Finding 3.6     |
| 7 | Discover &mdash; firewall-drop block event          | Finding 3.7     |
| 8 | Endpoints view &mdash; agent status/OS/groups       | Section 4       |

## 12. Analyst Reflection

Building this investigation helped me understand the difference between generating telemetry and actually investigating it. Correlating account-change events across a single host, and separately correlating a brute-force sequence across host and attacker, showed me how much more useful a timeline is than any single alert on its own. I also got hands-on practice confirming that a custom detection rule and its Active Response action both fired correctly end-to-end. This exercise reinforced the value of reviewing endpoint status (via Endpoints) alongside alert data, so that gaps in telemetry aren't mistaken for gaps in attacker activity.

## 13. Investigation Limitations

- Exact operator intent behind the Windows account changes cannot be determined from telemetry alone.
- The lab activity was intentionally generated by me and does not represent an uncontrolled real-world compromise.
- Some timestamps are approximate where multiple closely-spaced events were grouped for readability.
- No source IP address for the Kali attacker VM was captured in the reviewed screenshots; it is identified by role (attacker VM) rather than by address in this report.

## 14. Final Quality Check

- All findings are supported by Wazuh Discover/Dashboard evidence.
- Event IDs and rule descriptions match what was observed in the lab.
- Hostnames are consistent throughout (MYDFIR-Saqo, saqo-linux).
- The timeline is chronological.
- SSH abuse is supported by multiple correlated events, not a single failed login.
- Active Response is supported by an actual logged event.
- The report distinguishes lab simulation from production interpretation.
- Recommendations directly address the findings.
- No passwords, tokens, or unnecessary sensitive information are included.